

Anil Gogebakan

Research Statement

My research focuses on productivity, growth, international trade, and environmental economics. I study how changes in trade, economic policy, and energy markets affect productivity, production patterns, and the distribution of economic and environmental outcomes across industries, regions, and countries. My research combines empirical analysis with quantitative economic models and detailed economic and spatial data. A common theme is understanding how changes in markets and economic policy affect aggregate productivity and growth through the input allocation across industries and regions. My current work spans international trade, productivity, and environmental policy, while my future research will increasingly connect these areas by studying how trade and environmental policies interact and how the energy transition affects productivity and economic activity.

Current Research

My published research examines the relationship between international trade and income. In *Revisiting Trade and Income in the New Era of Globalisation: Distance, Big Boats and Natural Barriers to Trade*, published in *The World Economy*, I develop a new instrumental-variable strategy to estimate the causal effect of trade on income. The identification strategy exploits changes in effective maritime distance generated by the interaction between growing container ships and capacity constraints associated with the Panama Canal. The results provide new evidence on how trade affects economic activity and growth.

My job market paper, *Energy Misallocation and Aggregate Productivity: Evidence from Canada*, examines the role of energy misallocation in aggregate productivity. I extend the standard misallocation framework to include energy alongside capital and labor and use Canadian provincial input-output data to quantify productivity losses across industries and provinces. While the existing misallocation literature has largely focused on capital and labor, energy is also a fundamental input into production. I find that eliminating all input misallocation could increase aggregate productivity by 5 to 8 percent. Interestingly, energy contributes disproportionately to these productivity losses relative to its share of production costs. Despite its relatively small cost share, its contribution to aggregate productivity losses is comparable to capital and larger than labor. These findings highlight an underexplored channel through which energy markets can affect aggregate productivity and motivate broader questions about how energy prices, infrastructure, and policy affect productivity during the transition to cleaner energy.

A third strand examines environmental policy and its distributional consequences. In *Uneven Gains: The Environmental Justice Impact of the Regional Greenhouse Gas Initiative (RGGI) on Local Air Pollution and Health*, we combine power-plant emissions data with a heterogeneous difference-in-differences design and the InMAP reduced-complexity air pollution model to study how RGGI affected local pollution exposure and health outcomes. We find that the environmental benefits of the policy were asymmetrically distributed across regions and communities.

Future Research Agenda

My future research will develop these strands into a broader program on productivity, trade, and environmental policy. A central question is how policies affecting production and trade interact with environmental policy and the transition to cleaner energy, and how these changes affect aggregate productivity, growth, and their distribution across industries, regions, and countries.

First, I plan to study the trade-environment nexus. International trade changes the location of production and allows environmental policies in one country to affect production and emissions elsewhere. The recent rise in tariffs and other trade barriers makes these questions particularly relevant. Changes in trade costs can alter where goods are produced, the composition of trade, and global value chains, with potentially important environmental consequences. I am particularly interested in how carbon pricing, carbon border adjustment mechanisms, environmental standards, and renewable-energy policies affect comparative advantage, trade patterns, and global value chains. This research will examine whether differences in environmental policy alter the location of production, particularly in energy-intensive industries, and whether rising trade barriers shift production toward locations with different environmental and energy characteristics.

Second, I plan to continue studying productivity and energy markets. My job market paper shows that differences in the use and cost of energy can have substantial consequences for aggregate productivity, with energy contributing disproportionately to productivity losses relative to its input share. I plan to investigate mechanisms including energy prices, infrastructure constraints, and access to electricity. The expansion of renewable generation and electrification creates important questions about how changes in energy markets affect productivity and investment across regions and industries. I am particularly interested in whether investments supporting decarbonization can also improve productivity by reducing differences in energy costs and improving access to energy infrastructure. This work will connect the energy transition to the macroeconomic literature on productivity and misallocation.

Third, I will examine the local and distributional effects of environmental and energy policies. Building on my RGGI research, I plan to study how changes in energy production, environmental regulation, and cleaner technologies affect pollution exposure, economic activity, and health across regions and communities. Detailed spatial data can identify not only aggregate effects but also who benefits and who bears the costs of environmental policies.

Finally, I plan to continue developing my research on trade, productivity, and growth. My published work examines how changes in trade costs affect income, while my productivity research studies differences in production conditions across regions and industries. I plan to connect these perspectives by studying how trade and changes in market access affect productivity and economic activity across countries and industries.

Research Approach

My research combines empirical analysis, quantitative economic models, and detailed economic and spatial data. I use instrumental variables, difference-in-differences, and related empirical approaches to identify causal relationships, complemented by quantitative models when useful for understanding mechanisms and evaluating policy counterfactuals. My work has involved international trade data, provincial and industry-level input-output data, power-plant emissions, and high-resolution spatial and demographic data. I work primarily with Python and Stata and have additional experience with InMAP and spatial data analysis.

In summary, my research develops a coherent program at the intersection of productivity, international trade, and environmental economics. My agenda is centered on understanding how changes in trade, energy markets, and economic policy affect productivity and growth, and how environmental policy and the energy transition alter these relationships.